BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 201 · Hash tables and collisions

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. In an undirected graph, why does edge A–B appear in the adjacency list of both A and B?

Q2. In a BST, 18 is smaller than root 25. Which direction do you move?

Q3. What are valid indexes of a five-element C# array?

Q4. Calculate 17 MOD 6.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

hash function — Maps a key to an initial bucket index.

collision — Two keys map to the same initial bucket.

linear probing — Checks successive buckets, wrapping around when needed.

load factor — The proportion of occupied table buckets.

clustering — A sequence of occupied buckets increasing probe lengths.

New learning

Calculate the index — 9 MOD 7 = 2, so put 9 into bucket 2.

Detect a collision — 16 MOD 7 is also 2. The bucket is occupied, so move to the next bucket, 3.

Probe until empty — 23 also hashes to 2; buckets 2 and 3 are full, so place it in 4.

Worked example

Insert 6, 11, 16 into five buckets using h(k)=k MOD 5 and linear probing.

1. Predict and explain — Insert 6 at index 1, because 6 MOD 5 = 1.

2. Check and extend — 11 collides at 1, so store it at 2.

3. Check and extend — 16 collides at 1 and 2, so store it at 3.

We Do · Practise with guidance

Insert 8, 13, 18 into five buckets using h(k)=k MOD 5.

```
8 MOD 5 = 3; 13 MOD 5 = 3; 18 MOD 5 = 3
```

Predict and explain

8 goes into bucket 3.

Check and extend

13 collides at 3 and probes to 4.

18 collides at 3 and 4; linear probing wraps around to 0, which is empty, so 18 goes into bucket 0.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Using h(k)=k MOD 7, find the starting index for 24.

Question 2

Insert 4, 9, 14 into five empty buckets using h(k)=k MOD 5 and linear probing. Give final indexes.

Question 3

Explain the effect of a high load factor on linear probing.

Exit Ticket · Answer before checking

A hash collision means two different keys have the same starting bucket.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Because traversal is permitted both ways.

Q2. Left.

Q3. 0, 1, 2, 3, 4.

Q4. 5, since 17 = 2×6 + 5.

You Do pages · Independent questions

Q1. 24 MOD 7 = 3.

Q2. All hash to 4. Insert 4 at index 4; 9 probes to 0; 14 probes past occupied 4 and 0 to index 1.

Q3. More occupied buckets lead to longer probe sequences, clustering and slower expected lookup and insertion.

Exit Ticket · Model answer

True. A collision requires a resolution strategy such as linear probing.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: